

KOZASAJ, Kazakhstan

WMO: 355290

Lat: 48.217N

Lon: 57.117E

Elev: 495

StdP: 14.43

Time Zone: 5.00 (E05)

Period: 86-94

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-19.7	-15.1	-26.6	1.3	-19.5	-21.2	1.7	-14.7	36.1	14.2	31.3	14.7	1.5	50	0.581

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.5	96.1	69.7	92.7	68.1	89.4	67.1	73.4	91.3	71.0	87.6	68.8	85.0	6.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.3	102.4	81.9	65.2	95.0	79.7	62.7	86.7	77.7	37.4	90.3	35.2	88.0	33.2	84.9	80.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min		Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
26.6	22.5	20.2		-25.6	102.4	3.9	2.1	-28.4	103.9	-30.7	105.1	-32.8	106.3	-35.6	107.9	
				-25.8	76.0	3.8	3.1	-28.6	78.2	-30.8	80.1	-33.0	81.9	-35.7	84.1	
				DB												
				WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.0	12.6	8.1	21.4	47.0	60.3	72.4	74.8	71.1	59.6	43.8	26.5	16.4	(6)
	DBStd	25.78	13.05	13.08	11.53	9.63	7.77	7.32	6.09	6.69	8.38	8.34	11.51	11.45	(7)
	HDD50	5389	1159	1173	887	167	12	0	0	0	20	226	705	1040	(8)
	HDD65	8847	1624	1593	1351	541	185	15	3	22	196	657	1155	1505	(9)
	CDD50	2848	0	0	0	78	331	673	769	655	308	34	0	0	(10)
	CDD65	834	0	0	0	2	39	238	308	212	35	0	0	0	(11)
	CDH74	10354	0	0	0	43	535	3121	3707	2427	512	9	0	0	(12)
	CDH80	4536	0	0	0	6	121	1523	1753	1002	131	0	0	0	(13)
	WSAvg	7.0	7.7	8.3	8.2	8.3	7.7	6.4	6.1	6.2	6.3	6.1	6.9	6.4	(14)
	PrecAvg	14.5	1.3	1.2	1.5	1.3	1.5	1.1	1.1	0.6	0.6	1.0	1.5	1.7	(15)
(16) Precipitation	PrecMax	23.3	2.6	2.8	4.4	3.7	3.9	4.8	3.7	1.7	2.3	2.0	3.2	4.1	(16)
	PrecMin	8.0	0.1	0.0	0.0	0.3	0.1	0.0	0.0	0.0	0.1	0.0	0.0	0.4	(17)
	PrecStd	4.0	0.7	0.7	1.0	0.8	1.0	1.0	0.8	0.4	0.5	0.5	0.7	1.0	(18)
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.0	36.8	55.6	79.1	86.6	101.4	99.8	95.8	88.0	75.3	49.2	38.2	(19)
		MCWB	33.2	33.9	46.1	58.4	64.0	74.4	72.8	66.3	67.9	58.0	42.0	36.9	(20)
	2%	DB	33.4	33.4	46.3	73.2	83.6	96.9	96.2	92.2	83.8	69.6	44.9	35.5	(21)
		MCWB	31.8	32.1	40.5	54.5	62.1	69.8	71.2	66.6	63.8	52.4	39.6	34.5	(22)
	5%	DB	31.8	31.1	40.9	68.2	80.0	92.7	92.9	89.1	79.7	63.2	42.2	33.4	(23)
		MCWB	30.4	29.9	37.0	52.1	59.4	67.5	69.5	65.9	60.7	49.3	37.5	32.0	(24)
	10%	DB	29.7	27.0	36.9	63.9	76.5	88.9	89.2	86.0	76.1	58.2	39.3	31.8	(25)
		MCWB	28.4	25.5	33.8	50.3	58.1	66.3	68.3	65.2	59.0	47.0	35.2	30.3	(26)
	0.4%	WB	33.5	34.3	46.3	60.0	67.1	76.0	77.5	73.6	69.1	58.3	44.3	37.1	(27)
		MCDB	34.7	36.0	53.9	74.5	83.7	98.7	93.9	87.6	87.2	71.7	46.2	37.6	(28)
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	32.0	32.3	40.8	56.8	63.3	72.4	74.3	70.4	65.2	54.7	41.1	34.7	(29)
		MCDB	33.3	33.4	45.1	70.8	79.9	92.8	92.3	85.1	80.6	65.3	43.8	35.4	(30)
	5%	WB	30.6	29.9	37.0	53.9	61.5	69.9	72.0	68.1	62.5	51.3	38.1	32.6	(31)
		MCDB	31.7	31.3	40.4	66.2	76.4	88.1	87.9	83.9	75.4	60.9	41.1	33.5	(32)
	10%	WB	28.6	25.4	34.1	51.3	59.3	67.7	69.5	66.2	60.1	47.9	35.7	30.5	(33)
		MCDB	29.7	26.9	36.7	62.5	73.0	83.4	85.2	82.2	73.8	56.4	38.5	31.7	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	13.4	15.4	16.2	20.0	22.1	24.4	23.5	23.6	24.0	21.7	13.2	13.1	(35)
		MCDBR	11.4	10.8	16.5	26.7	27.0	29.1	27.7	27.3	30.0	30.4	17.5	10.9	(36)
		MCWBR	11.2	10.4	11.0	13.7	12.9	13.6	13.6	12.5	16.8	17.8	13.1	10.4	(37)
	5% WB	MCDBR	11.0	10.9	15.1	24.3	24.6	26.8	25.5	24.6	27.4	27.0	14.4	10.4	(38)
		MCWBR	10.9	10.5	10.4	13.3	13.5	14.6	14.3	13.4	16.9	17.4	11.8	10.0	(39)
(40) Clear-Sky Solar Irradiance	taub		0.310	0.336	0.402	0.445	0.435	0.429	0.421	0.425	0.373	0.366	0.343	0.317	(40)
	taud		2.089	2.015	2.003	2.021	2.130	2.206	2.247	2.210	2.307	2.240	2.207	2.119	(41)
	Ebn at Noon		228	250	252	254	263	265	266	258	262	240	216	207	(42)
	Edh at Noon		28	38	46	50	47	44	42	41	34	31	25	24	(43)
(44) All-Sky Solar Radiation	RadAvg		452	820	1154	1538	1966	2184	2108	1856	1407	824	410	290	(44)
	RadStd		79	86	123	157	158	150	120	96	84	92	48	52	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (3 neighbors)	+1.17	N/A	N/A	+2.27	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page